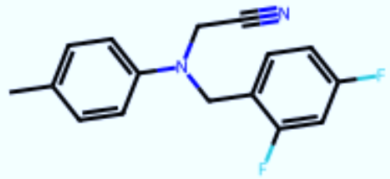
DSA103 Advanced Chemical Data Science

Lecture 15: Data wrangling without loss

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

	Smiles	Standard Type	Standard Relation	Standard Value	Standard Units	Assay ChEMBL ID	ROMol
...					g.mL-1	CHEMBL3625335	
	@H]1CC[C@@H]2CN1C(=O)N2OS(=O)(=O)[O-].[Na+]	MIC	'<='	0.120	ug.mL-1	CHEMBL5247824	
							

Assignment_Session12: A new antibiotic

- Discussion of solution
- Q&A

Data wrangling – outline and recap

What is data wrangling?

- Process of cleaning, restructuring, and enriching raw data into a usable format
- Essential before analysis, visualisation, or modeling

Why?

- Improved data quality
- Increased efficiency
- Better decision-making

Data wrangling without loss





- Cleaning, transforming, and reshaping safely
- Every record and value is preserved
- Operations are reversible

Data wrangling – “types”

Cleaning:

- Fix typos
- Convert types
- Fill NAs

No data loss

Transformation:

- Renaming columns
- Pivoting
- Merging

No data loss (if careful)

Reduction:

- Filter
- Aggregate
- Drop columns

Data loss

Activity: Working on a messy dataset

Inspect the “patients_results_messy.csv” dataset using the data wrangling notebook.

1) What is wrong with the dataset?
   - Inconsistent labelling
   - Inconsistent casing
   - Missing values
   - Mixed data types
   - …

2) Fix the issues without loosing any data!

3) Discuss the reversibility of your steps and how you can make sure that you did not loose any data

Pivoting and melting

Opposite operations for reshaping data frames

Melt():

- Convert wide to long (column headers into rows)
- Converts Dataframe to a format of identifier and measured variables
- Common use: Data cleaning, preparation for plotting
- Flexible, handling of duplicates supported

Pivot():

- Convert long to wide (a column with multiple values into several columns of their own)
- Common use: Data summarisation
- No handling of duplicates (index!)
- pivot_table() tolerates duplicates, but performs aggregation

https://jakevdp.github.io/PythonDataScienceHandbook/03.09-pivot-tables.html
https://pandas.pydata.org/docs/user_guide/reshaping.html

Melt

df3

	first	last	height	weight
0	John	Doe	5.5	130
1	Mary	Bo	6.0	150



```
df3.melt(id_vars=['first', 'last'])
```

	first	last	variable	value
0	John	Doe	height	5.5
1	Mary	Bo	height	6.0
2	John	Doe	weight	130
3	Mary	Bo	weight	150

Pivot

```
df.pivot(index='foo',
         columns='bar',
         values='baz')
```

df

	foo	bar	baz	zoo
0	one	A	1	x
1	one	B	2	y
2	one	C	3	z
3	two	A	4	q
4	two	B	5	w
5	two	C	6	t

Stacked



bar	A	B	C
foo			
one	1	2	3
two	4	5	6

Record

Activity: Pivot / melt

1) Demo of pivot and melt

2) Discuss the benefit of a long-table vs. a wide table.

3) Play through the exercise on the iris dataset.

Joining datasets – concat():

Always check shape before and after transformations!

pd.concat():

- Like **np.concatenate()**, also allowing specification of axis (default: rows)
- By default: preservation of indices (**ignore_index=False**)
- Data source can be added by **keys** option
- Tracking duplication: flag **verify_integrity** raises exception
- Joins can specify the type of concatenation (e.g. inner vs. outer)
- **Append()** is a method of Pandas objects as a shorthand for concat

```
try:
    pd.concat([x, y], verify_integrity=True)
except ValueError as e:
    print("ValueError:", e)
```

```
ValueError: Indexes have overlapping values: [0, 1]
```

https://pandas.pydata.org/docs/user_guide/merging.html

Joining datasets – merge():

Always check shape before and after transformations!

merge(): Combines two DataFrames based on common columns or indices

Key parameters:

- **left, right**: The two DataFrames to merge
- **on**: Column(s) to join on (must exist in both DataFrames)
- left_on, right_on: Specify different column names to join on if they differ
- **how**: Type of join
  - **'inner'** → only matching rows (default)
  - 'left' → all rows from left + matches from right
  - 'right' → all rows from right + matches from left
  - 'outer' → all rows from both (union)
- Can also be called as method on the left dataframe

```
# Merge on canonical_smiles, how="inner" drops all rows which are only contained in one df
merged = pd.merge(act_clean, desc, on="canonical_smiles", how="inner")
```

https://pandas.pydata.org/docs/user_guide/merging.html

Joining datasets – join():

Always check shape before and after transformations!

join(): Combines two DataFrames **by their index** (or by a key column), similar to **merge()**

Typical use case: Join key = index

Key parameters:

- other: The DataFrame (or list of DataFrames) to join with
- **how**: Type of join — 'left', 'right', 'outer', 'inner' (default **'left'**)

```
left_df = pd.DataFrame(
    {"A": ["A0", "A1", "A2"], "B": ["B0", "B1", "B2"]}, index=["K0", "K1", "K2"]
    )

right_df = pd.DataFrame(
    {"C": ["C0", "C2", "C3"], "D": ["D0", "D2", "D3"]}, index=["K0", "K2", "K3"]
    )

result_df = left_df.join(right_df)
```

https://pandas.pydata.org/docs/user_guide/merging.html

Activity: Merging dataframes

Merge the previous dataframe from the “patients_results_messy.csv” with the “patients_info.csv”.